Code No: 115EM **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
SOFTWARE ENGINEERING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Compare and contrast waterfall model and spiral model.
b) Explain the Unified process. Defend importance of the unified process. [7+8]

2. What is software process? Appraise the need of software process improvement? Discuss capability maturity models. [15]

3.a) What are the various tasks in requirements engineering? Illustrate with an example the task of collecting requirements at various steps.
b) How a data flow model is created? Illustrate with an example. [8+7]

4.a) Illustrate on SRD (Software Requirements Document).
b) Explain Requirements elicitation and management. [7+8]

5.a) What is Software architecture? Analyse various architectural styles and patterns.
b) Illustrate with neat diagrams the process of mapping data – flow into a software architecture. [7+8]

6.a) Explain design process and various design quality guidelines.
b) Explain various types of cohesion and coupling. List in the order of level of cohesion and coupling preferred for component level design. [7+8]

7.a) Explain the strategic approaches to software testing. Why testing is difficult? Elaborate on how to overcome.
b) Give an overview of black-box testing strategy. Explain Equivalence partitioning and boundary value analysis. [7+8]

8.a) What is software quality? Explain various steps for software quality assurance.
b) What are various risks associated with software? Explain steps to mitigate monitor and manage risks. [7+8]

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